Jeongsik Park

Plano, TX | 945-217-6306 | lucas.jeongsik.park@gmail.com | linkedin.com/in/jeongsik-park | webpage

Education

University of Southern California M.S., Computer Science (AI Specialization) May 2026
The University of Texas at Dallas B.S., Computer Science (GPA: 3.94) May 2024

Technical Skills

AI Engineering Python, PyTorch, LLM/VLM, Agentic Workflows (LangGraph, tool/function calling, memory, MCP), RAG (LlamaIndex, pgvector, hybrid search, reranking), vLLM
Backend & Data FastAPI, REST APIs, Pydantic, PostgreSQL
Cloud & MLOps AWS (Bedrock, ECS Fargate, EC2, RDS, S3), Terraform, Docker, Kubernetes, Helm, CI/CD
Observability OpenTelemetry, LangSmith, CloudWatch, Prometheus, Grafana

Professional Experience

IntelliChoice July 2026 – Present
AI Engineer (Remote) Lewisville, TX

- Built and deployed a production-grade AI education platform with two FastAPI/React applications serving **1,200+ K–12 students**; developed REST APIs backed by PostgreSQL and deployed containerized services on AWS ECS Fargate with RDS, Terraform, and GitHub Actions CI/CD.
- Engineered a security-first RAG system with metadata-scoped retrieval, citation-gated fail-closed responses, and LlamaIndex hybrid search combining HNSW-indexed pgvector with PostgreSQL full-text search via reciprocal rank fusion and LLM reranking.
- Built a centralized AWS Bedrock model gateway with a **23-task registry** for task-specific model routing and budget controls; implemented LangGraph agentic workflows with Pydantic-validated tool/function calling, MCP integrations, and human-in-the-loop approval for **100% of external side-effecting actions**.
- Architected a two-tier LLM memory system that consolidates append-only learning events into long-term semantic memories to support personalized learning, bounded at 20K input tokens and 4 model calls per student; enforced evidence provenance and a four-state lifecycle for resolving contradictory memories.
- Built an AI question-generation pipeline with SymPy validation, embedding-based deduplication, two blind solver models, and an independent LLM-as-a-Judge, scaling to **471 approved items across 112 skills** with **127/127 solver–author agreement and 12/12 negative-control detection**.
- Developed an LLM evaluation and observability platform with a 68-case golden Q&A test set, merge-blocking CI evaluations, and live-model runs achieving a **97.3% correct-refusal rate and 6/6 adversarial containment**, with OpenTelemetry instrumentation, PII-redacted LangSmith traces, and CloudWatch monitoring.

GE HealthCare May 2025 – May 2026
AI Engineer Intern (Hybrid) Bellevue, WA

- Built a LangGraph-based agent orchestration workflow with typed state, deterministic routing, and structured agent interactions; demonstrated the system to **clinicians from leading medical institutions at RSNA 2025**.
- Fine-tuned a Qwen3.5-35B-A3B MoE model for radiographic-positioning video classification using multi-GPU training with DeepSpeed ZeRO-3 Offload and FlashAttention-2, improving accuracy by **28%** over the vanilla baseline and serving the model with vLLM at sub-second latency — **co-inventor on a filed U.S. patent application**.
- Developed an iterative self-refinement method for VLM generation, reducing generation errors by **80%**.
- Built a SAM2-assisted annotation tool to accelerate labeling of ∼8,000 training images by **6×**; fine-tuned keypoint detection models, improving validation accuracy by **25%** for automated machine positioning.
- Benchmarked GPT, Opus, and MedGemma for radiology report generation using DICOM/NRRD medical imaging across MRI, mammography, and X-ray, establishing evaluation baselines.
- Deployed and optimized an NVIDIA digital human platform on AWS g5.12xlarge, consolidating the runtime from **2 GPUs to 1 GPU** and co-deploying TensorRT-optimized ASR/TTS models using Docker, Kubernetes, and Helm.
- Built a React/Flask dashboard streaming LangGraph state from Redis for real-time debugging for a **7-engineer team**.

Research Experience

Human Language Technology Research Institute August 2022 – May 2025
Undergraduate NLP Researcher (Advisor: Dr. Vincent Ng) Richardson, TX

- Led creation of the first unified meme-understanding dataset, achieving state-of-the-art results across interpretation, explanation, and categorization; *MemeInterpret* — **first author, Findings of EMNLP 2025**. [Link]
- Introduced intent description generation for memes using cultural background knowledge and few-shot in-context learning, improving LLM/VLM interpretation by 43%; *MemeIntent* — **first author, SIGDIAL 2024**. [Link]
- Developed topic-robust active learning for hate-speech detection, cutting annotation cost by **90%** while outperforming SOTA baselines; paper under review.